

Lesson 5-4 · Period 3

Radical Equations — Assessment-Critical:
Isolate a Variable + Table Completion

Klimsara-adapted · Student-centered · No Blooket

Do Now — Extraneous Solutions (HOT) Algebra 2

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DOK 3

5 min

bridge from P2

Higher Order Thinking · Extraneous Solutions

Some, but not all, equations with rational exponents have extraneous solutions.

Think: What is the relationship between the exponents and the possibility of having extraneous solutions? Explain your reasoning.

Predict: If you solved $V = \frac{1}{3}\pi r^2 h$ for r , would you expect an extraneous solution? Why or why not?

Launch — Rewrite a Formula + Table Completion

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DOK 2

12 min

teacher models Ex 2 + #44

Isolate-a-Variable Rules

1. Identify which variable to isolate.
2. Undo operations LAST-IN, FIRST-OUT: add/sub $\rightarrow$ mult/div $\rightarrow$ exponent/radical.
3. Raise both sides to the **RECIPROCAL** power to undo $x^{m/n}$: flip to $x^{n/m}$.

Table-Evaluation Rules

DOK 2

12 min

teacher models

Example 2 · Rewrite the Golden Gate for-**mula** Given: $x = \frac{\sqrt{y - 220}}{0.010583}$ Isolate y .

Steps: multiply both sides by coefficient $\rightarrow$ square both sides $\rightarrow$ add 220.
Then substitute $x = 200$ and evaluate y .

Practice #44 · Table Completion ASSESSMENT RE-HEARSAL — Item 2

Complete the table for $y = \sqrt[3]{2x + z} - 12$. Teacher walks through rows 1–2

Explore — Think-Pair-Share (35 min)

Algebra 2 .

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DOK 2

35 min

student work — TPS

6 items in order. Never cut #10 (Item 1A rehearsal). Cut #27 first if pace slows.

Try It 2 · Stopping distance formula $v = 3.57\sqrt{d}$

Rewrite to isolate d .

Practice #10 · Isolate **ASSESSMENT ITEM 1A RE-HEARSAL** Rewrite $y = \sqrt{(x - 48)/6}$ to isolate x .

Practice #25 / #26 / #27 · Rationalization

DOK 2

5 min

concept summary + assessment preview

Sentence Frame (#10 or #39) “To isolate ____, I first ____, then raise both sides to the power _____. The rewritten formula is _____. Plugging in $__ = __$ gives _____. In context this means _____.”

Topic 5 Assessment Preview Items 1A/1B — Rewrite a pyramid-volume formula for x , then evaluate. $\Rightarrow$ Rehearsed by **Practice #10**.

Item 2 — Complete the table from the rewritten formula. $\Rightarrow$ Rehearsed by **Practice #44** (done in Launch).

Exit Ticket — Pre-Assessment Confidence Check

Algebra 2 · Lesson 5-4 · Period 3

DOK 1-2

3 min

not graded

Complete on your exit ticket

1. To isolate a variable from a rational-exponent formula I _____.
2. When I evaluate the formula at a specific x , I _____.
3. One biggest thing I learned today: _____.